

Electronic Circuits II
Project: Delay

Students

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Date

July 2024

Completed during the second semester of 2024

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1 Introduction

The objective of this project was to design a delay pedal for electric guitar that processes the instrument signal and generates repetitions with adjustable time, number, and level.

The pedal includes controls that let the user customize its behavior: a repeats control for the number of repetitions, a time control for the interval between repetitions, a mix/level control for the proportion of original and processed signals, and a tone control for the tonal character of the repetitions.

The design is based on the PT2399 integrated circuit, a digital-analog device widely used in delay effects. It consists of five main stages:

1. **Power supply:** Provides the voltages required by the different stages.
2. **Input stage:** Includes DC blocking and a buffer.
3. **PT2399:** Handles the digital and analog processing of the delay.
4. **Tone control:** Adjusts the tonal character of the repetitions.
5. **Output summer:** Combines the processed and original signals.

The complete schematic is shown below. The colored areas identify the stages as follows: yellow (power supply), blue (input), large lower-left red area (PT2399), right red area (tone control), and green (output summer).

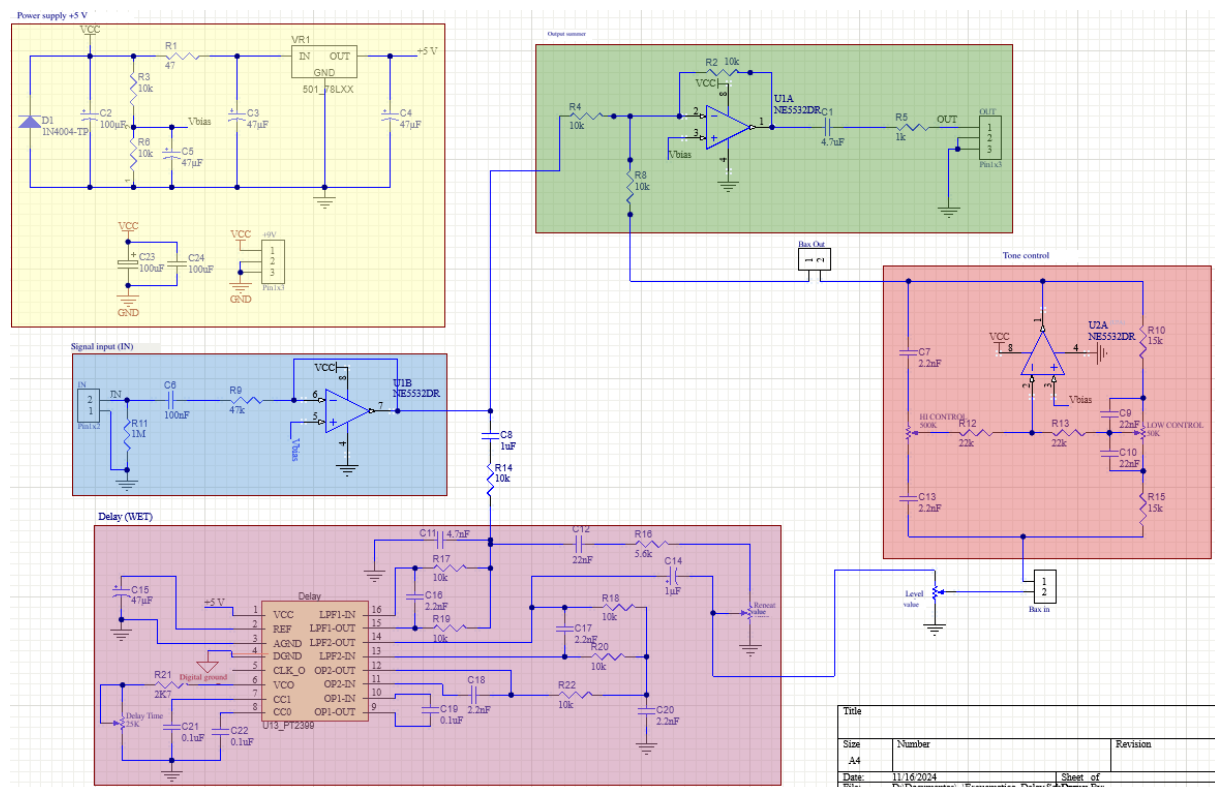


Figure 1: Complete circuit schematic and its five functional stages.

2 Power supply

The power supply provides different voltage levels for the circuit stages. The pedal takes an external 9 V supply and has no internal power source. The design generates the 4.5 V and 5 V levels required by the circuit from this supply:

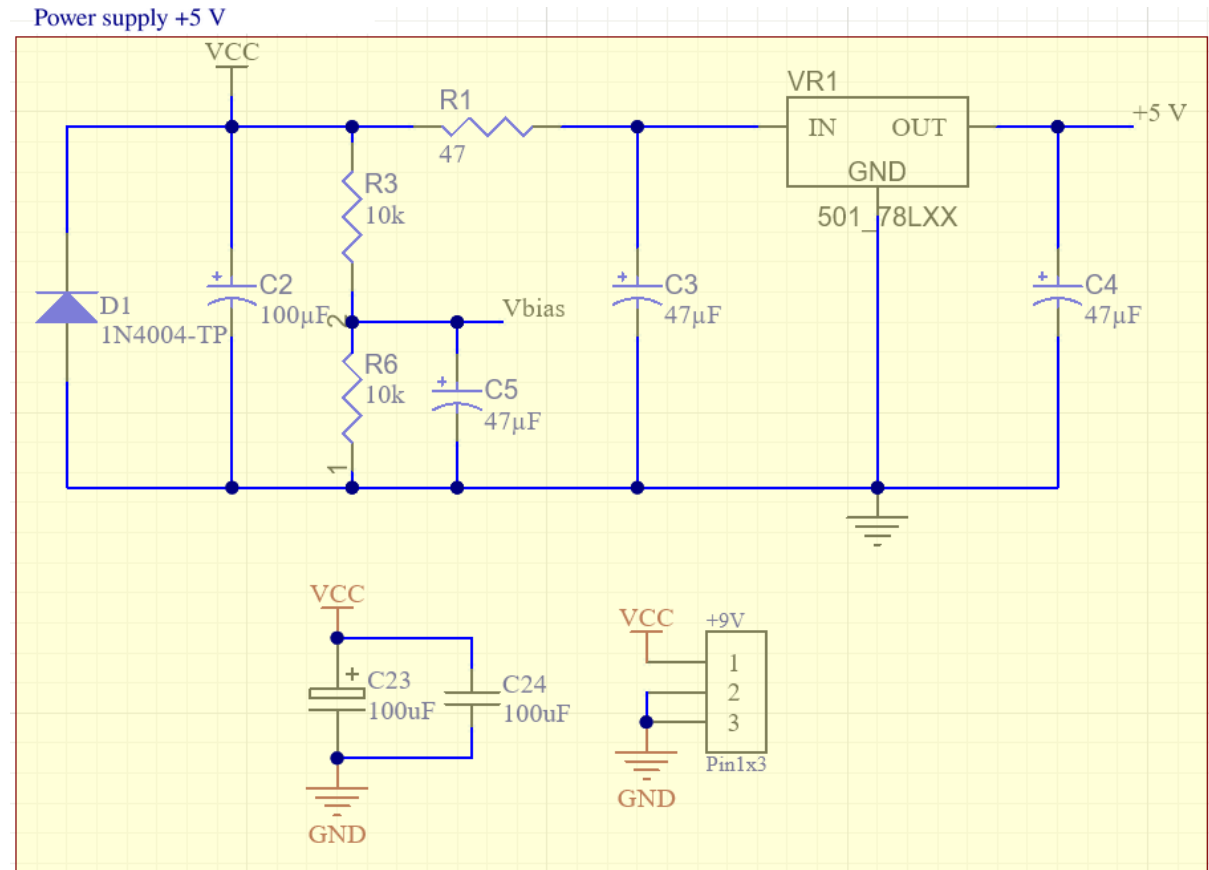


Figure 2: Power supply stage.

The 4.5 V level is a reference for the single-supply op-amps and biases the transistor buffer at the input. A divider comprising two 10 k Ω resistors produces this level, while a 47 μ F capacitor stabilizes it.

The PT2399 needs 5 V. A 7805 regulator supplies a stable, regulated voltage. An input diode protects the circuit against accidental reverse polarity.

3 Input stage

The pedal input uses an NE5532 op-amp on a single 9 V supply, with 4.5 V applied to its non-inverting input. Configured as a buffer, it presents a high input impedance, reducing the load on the preceding stage and preserving signal transfer:

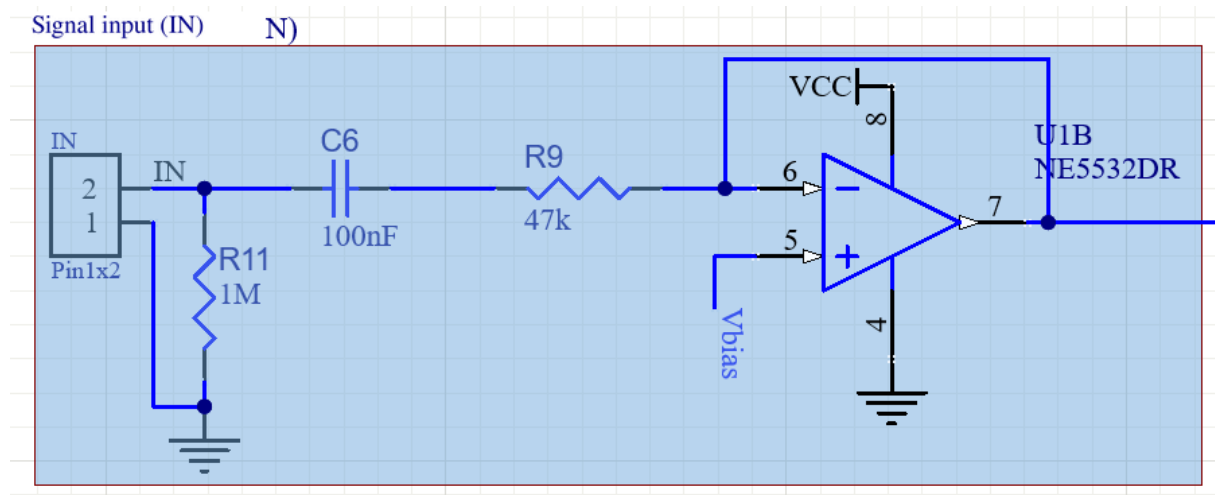


Figure 3: Input stage.

An input capacitor blocks any DC component from the device preceding the pedal. The 1 M Ω resistor to ground discharges the capacitor when that device is disconnected, helping avoid audible pops on connection or disconnection. Its high resistance also preserves the input impedance.

4 PT2399

According to the datasheet, the PT2399 can operate as a single delay or as an echo with multiple repetitions. The echo configuration uses the feedback loop shown in the following datasheet block diagram:

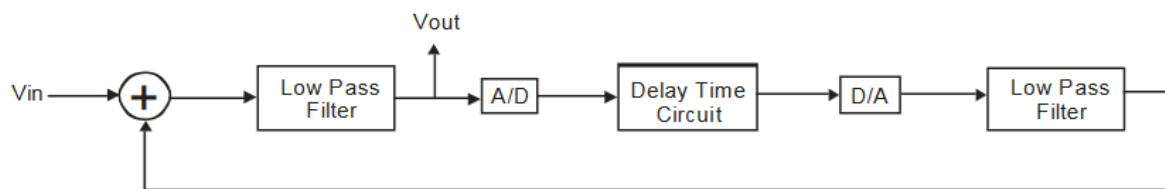


Figure 4: PT2399 echo-effect block diagram.

The PT2399 was configured following the datasheet recommendations, with some adjustments. Its ADC converts the input signal, which is then delayed digitally. Feeding part of the output back into the circuit generates multiple repetitions, as shown in the diagram and schematic. A potentiometer in this feedback loop sets the proportion of signal fed back, thereby controlling the number of repeats.

The implemented configuration is shown below:

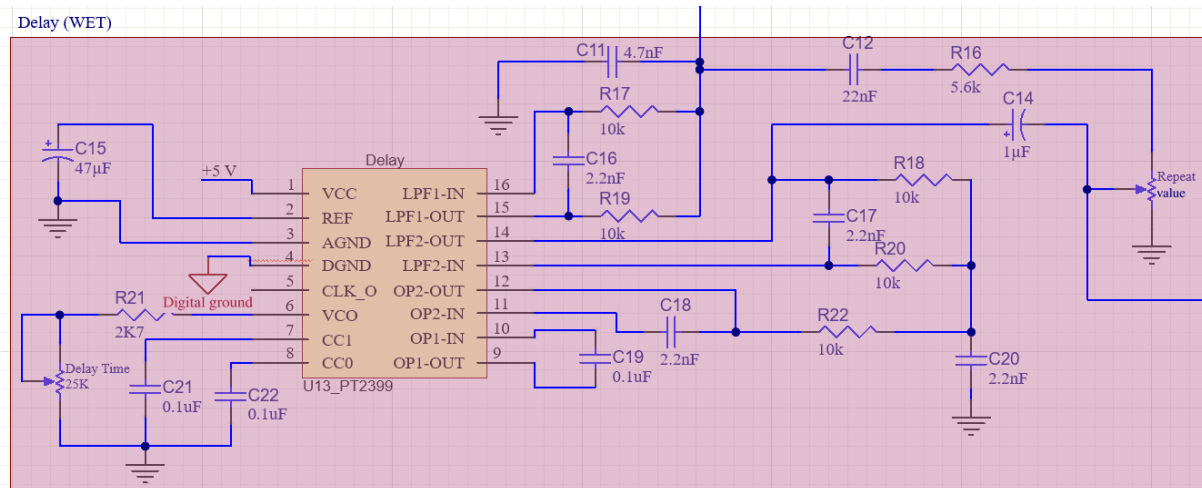


Figure 5: PT2399 stage.

The most relevant pins are pin 6 and pins 9–16. The remaining pins provide simple connections, including power, analog and digital grounds, and datasheet-specified components such as the capacitors on pins 7 and 8.

Analog-to-digital conversion takes place at the input on pin 16, so an anti-aliasing filter is needed. Pins 15 and 16 provide the inverting input and output of an internal op-amp for configuring this filter externally.

A multiple-feedback filter topology was selected. Its behavior was determined from the transfer function. Kirchhoff's current law was applied at the node connecting R15, R18, R12, and C8 (neglecting feedback), together with the inverting op-amp gain expression. Rearranging the result into the standard form of a second-order low-pass filter yields

$$\omega_0 = \frac{1}{\sqrt{R_{18}R_{15}C_8C_{13}}}. \quad (1)$$

The derivation is provided in the appendix. The filter was designed for a cutoff frequency of 4,949.48 Hz.

The same topology is used on the PT2399 output at pins 13 and 14 as a reconstruction filter to suppress high-frequency artifacts from the digital-to-analog conversion. Another partially configurable, first-order active low-pass reconstruction filter follows the DAC. Its resistor and capacitor values follow the datasheet.

The op-amp on pins 9 and 10, involved in analog-to-digital conversion, is also partially configurable. The recommended capacitor value is 100 nF, which was used here.

Finally, pin 6 controls the VCO. The resistance from this pin to ground determines the delay time. The datasheet provides a table of resistance values versus delay time, along with information such as distortion of the repetitions.

R	27.6K	21.3K	17.2K	14.3K	12.1K	10.5K	9.2K	8.2K
fck	2.0M	2.5M	3.0M	3.5M	4.0M	4.5M	5.0M	5.5M
td	342ms	273ms	228ms	196ms	171ms	151ms	136.6ms	124.1ms
THD	1.0%	0.8%	0.63%	0.53%	0.46%	0.41%	0.36%	0.33%

R	7.2K	6.4K	5.8K	5.4K	4.9K	4.5K	4K	3.4K
fck	6.0M	6.5M	7.0M	7.5M	8.0M	8.5M	9.0M	10M
td	113.7ms	104.3ms	97.1ms	92.2ms	86.3ms	81ms	75.9ms	68.1ms
THD	0.29%	0.27%	0.25%	0.25%	0.23%	0.22%	0.21%	0.19%

R	2.8K	2.4K	2K	1.67K	1.47K	1.28K	1.08K	894
fck	11M	12M	13M	14M	15M	16M	17M	18M
td	61.6ms	56.6ms	52.3ms	48.1ms	45.8ms	43ms	40.6ms	38.5ms
THD	0.18%	0.16%	0.15%	0.15%	0.15%	0.15%	0.14%	0.14%

R	723	519	288	0.5
fck	19M	20M	21M	22M
td	36.6ms	34.4ms	32.6ms	31.3ms
THD	0.14%	0.13%	0.13%	0.13%

Figure 6: Datasheet resistance values versus delay time.

A fixed minimum resistance prevents the total resistance from reaching zero when the control is turned down. The potentiometer and its series resistor were selected from this table.

5 Tone control

The tone-control design draws on engineer Ramón Vargas Patrón's analysis of an active tone-control network. Its topology and Bode diagram were used to understand the circuit:

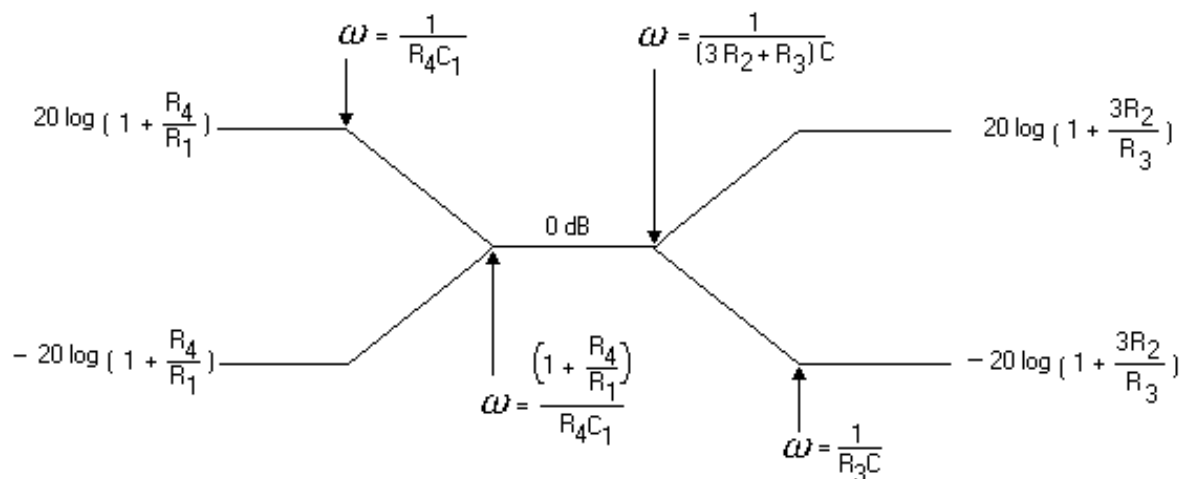


Figure 7: Tone-control network response for extreme positions of R and R4.

The target frequencies were 600 Hz for bass and 800 Hz for treble. The resulting values were 626.97 Hz and 822.08 Hz, with 12 dB of boost or cut. The following plots show the circuit's output voltage for a 1 V peak input; thus a 4 V output corresponds to 12 dB of gain.

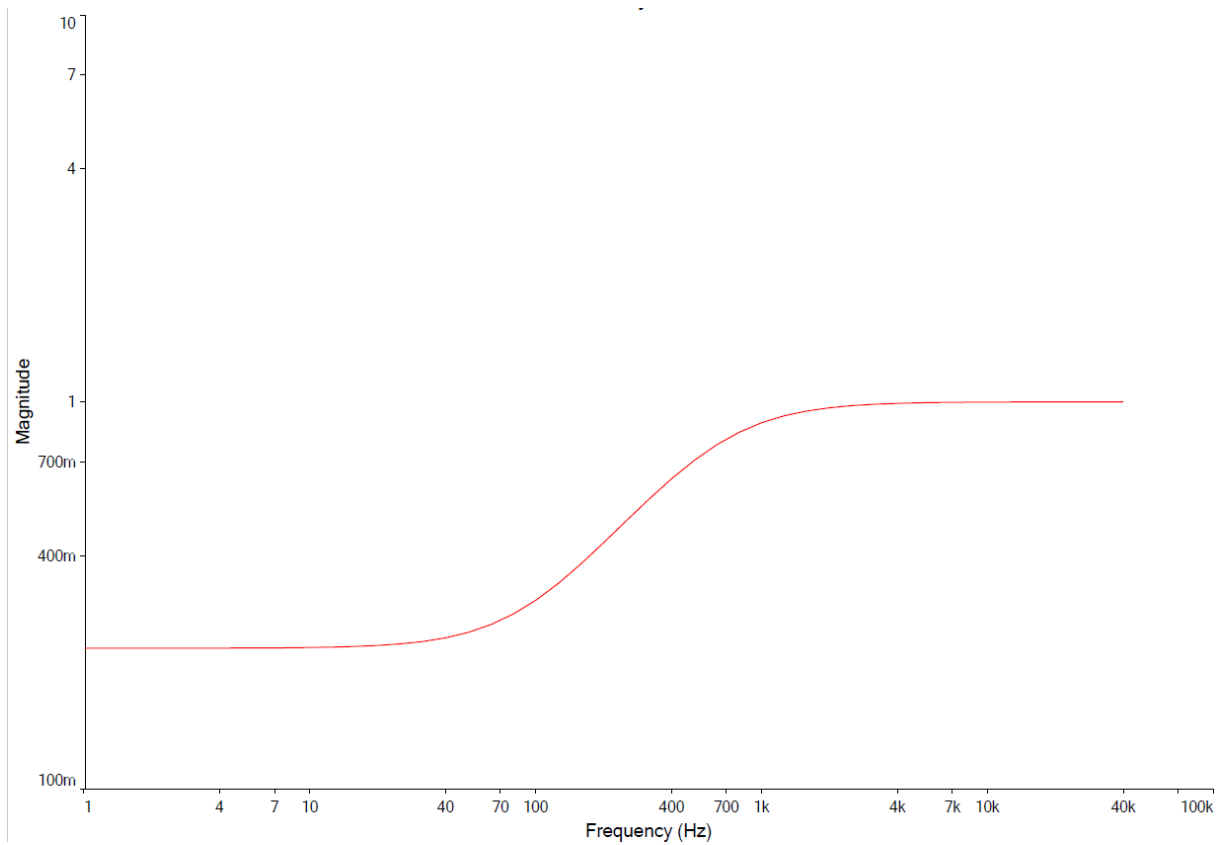


Figure 8: Maximum bass attenuation.

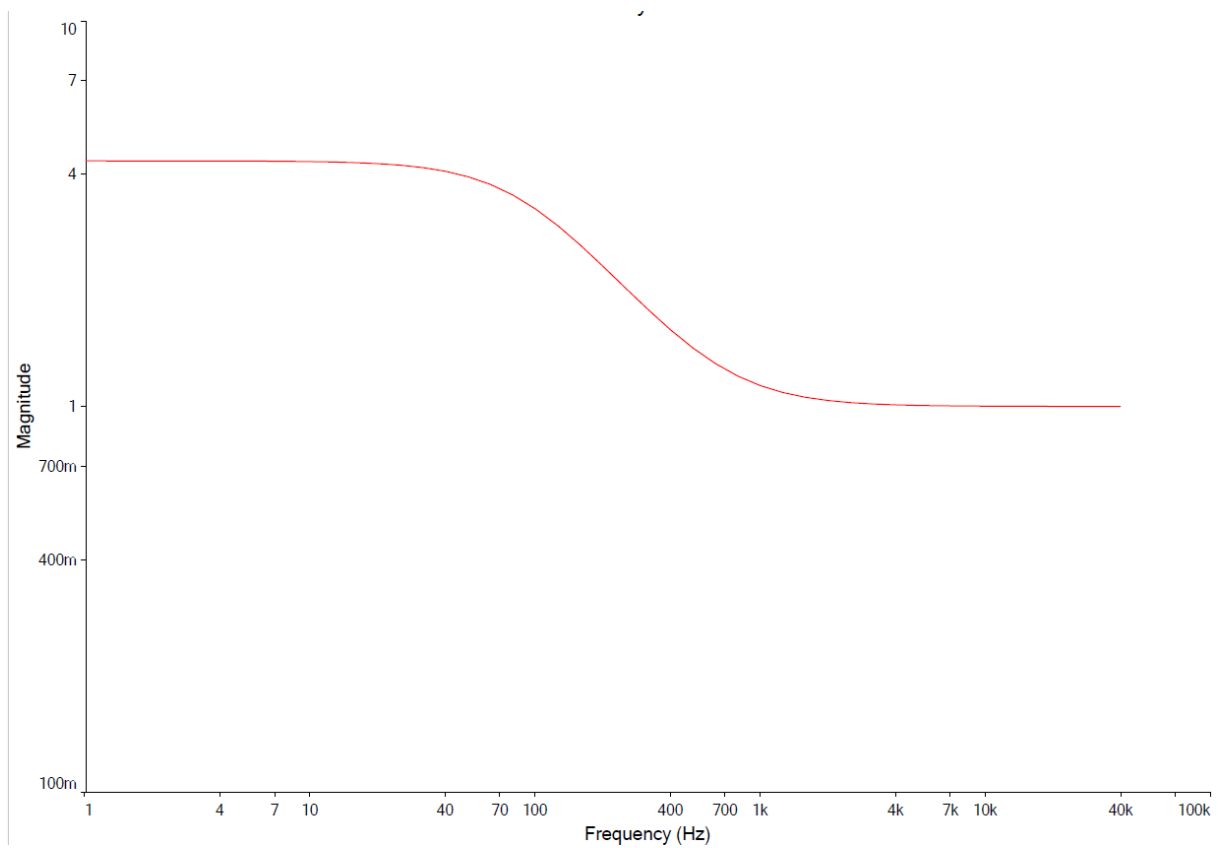


Figure 9: Maximum bass boost.

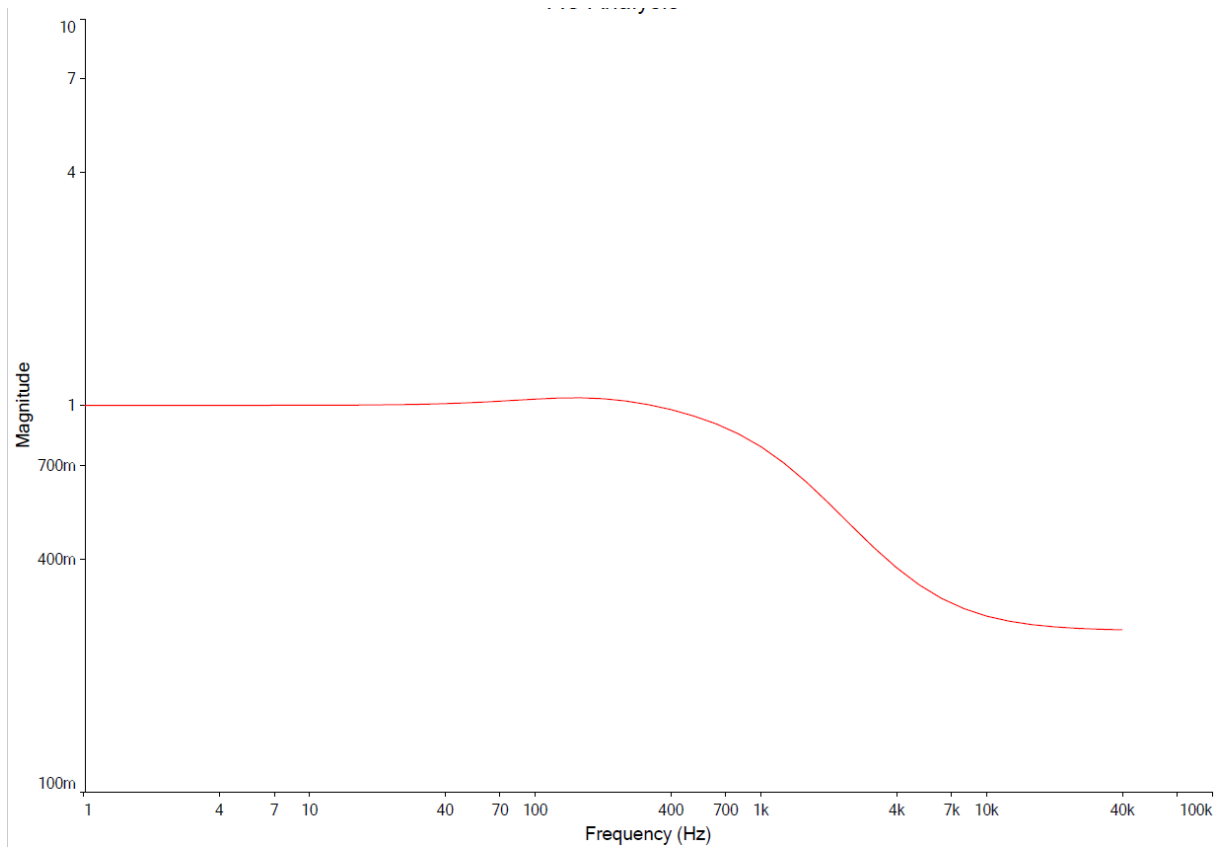


Figure 10: Maximum treble attenuation.

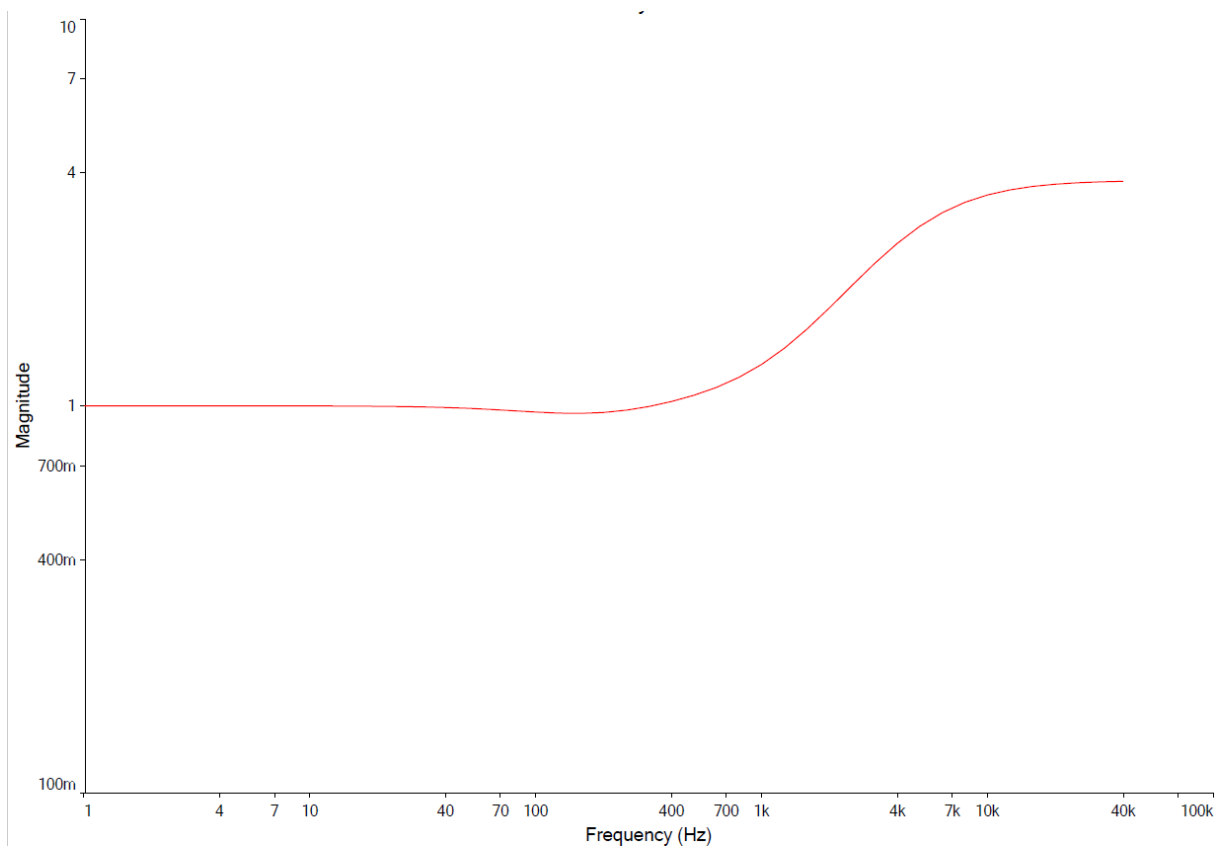


Figure 11: Maximum treble boost.

Two test points were added to the input and output of the tone-control circuit to evaluate its frequency response:

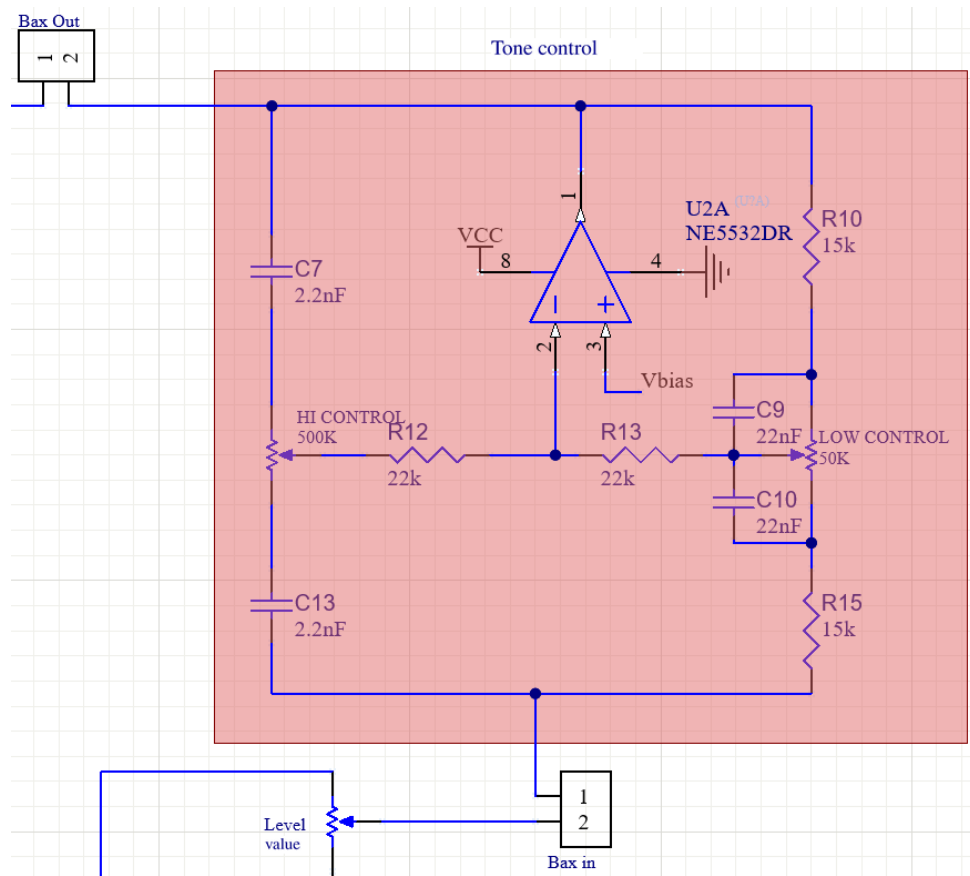


Figure 12: Tone-control stage with input and output test points (“Bax in” and “Bax Out” in the schematic).

6 Output stage

The output stage sums the original guitar signal and the processed delay signal. An op-amp summer with $10\text{ k}\Omega$ resistors combines them in equal proportions:

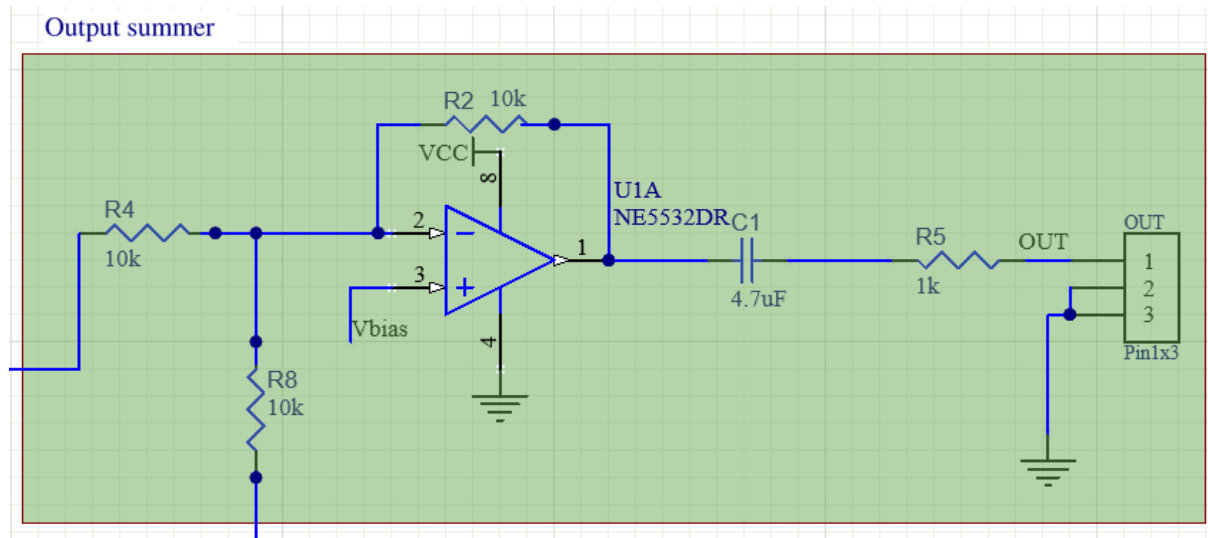


Figure 13: Output summing stage.

7 PCB

The PCB was designed for a guitar pedal. Surface-mount components were selected to keep it compact, resulting in a 5×5 cm board.

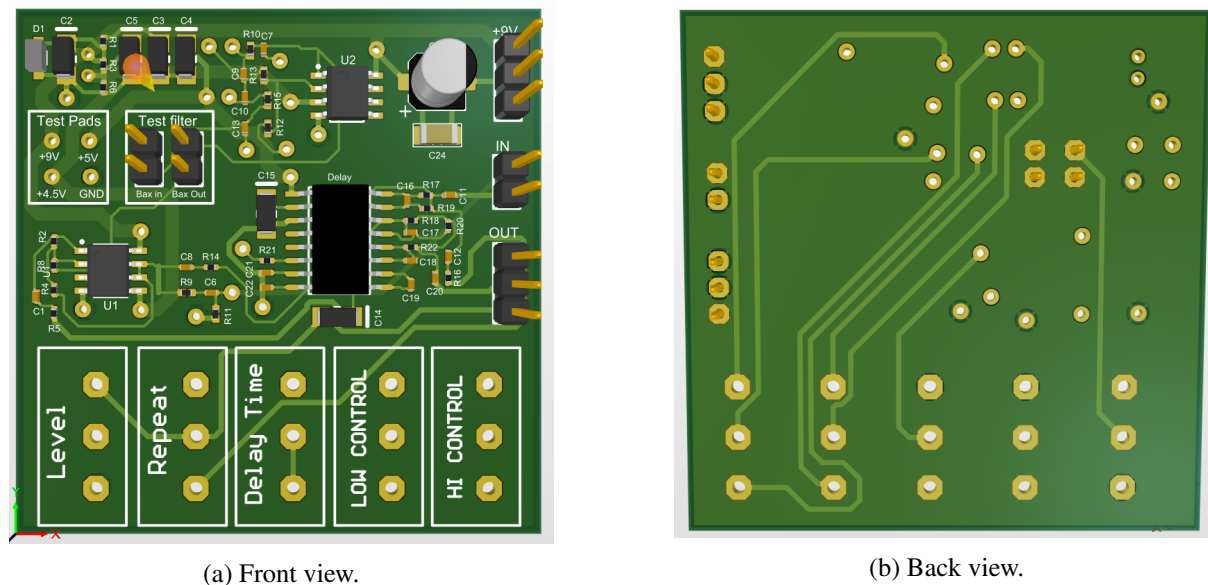


Figure 14: PCB.

The front includes test points for checking circuit voltages, as well as the input and output test points for the tone control described above.

The board has four layers:

- The top (red) and bottom (blue) layers route the component connections.
- One inner layer is a ground plane.
- The other inner layer (orange) distributes the supply voltages.

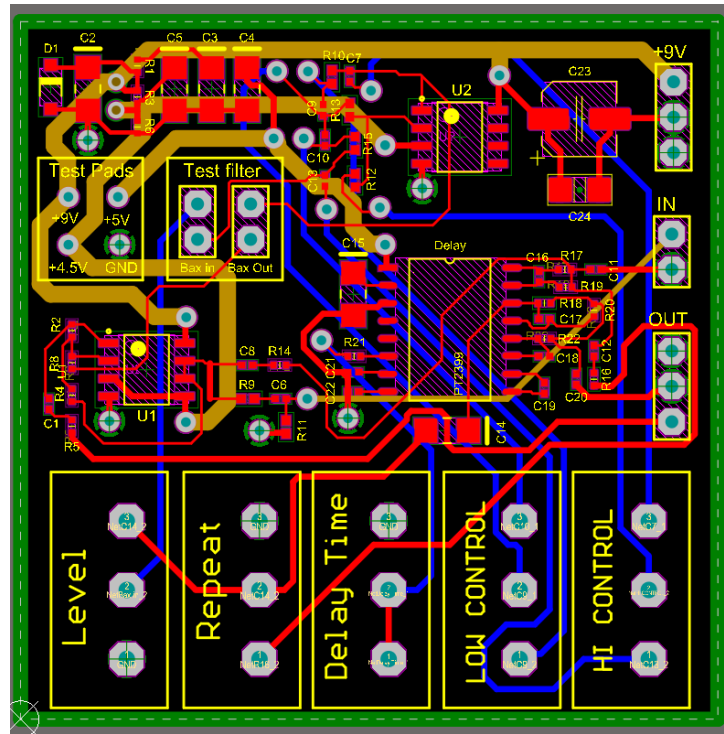


Figure 15: PCB layers in 2D.

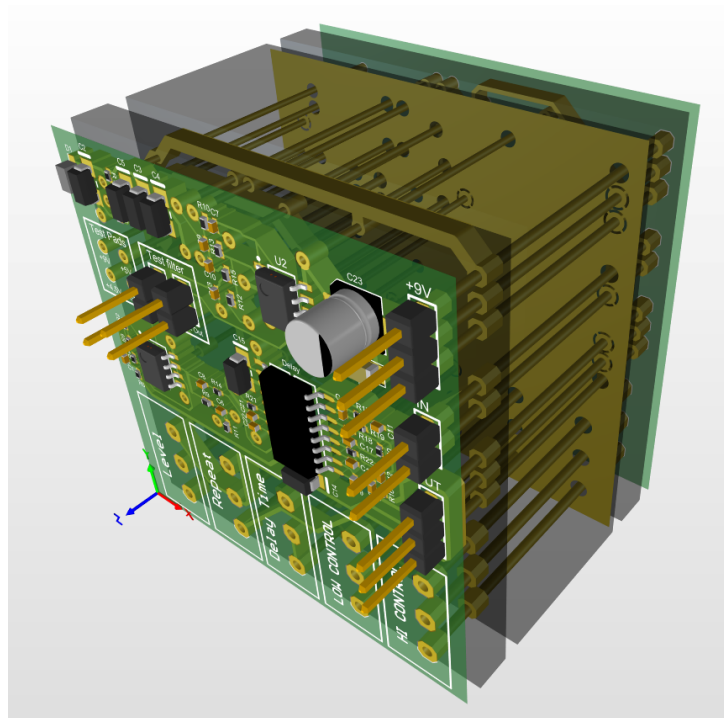


Figure 16: PCB layers in 3D.

8 Appendix

8.1 Transfer function of a multiple-feedback low-pass filter

The multiple-feedback low-pass topology is shown below:

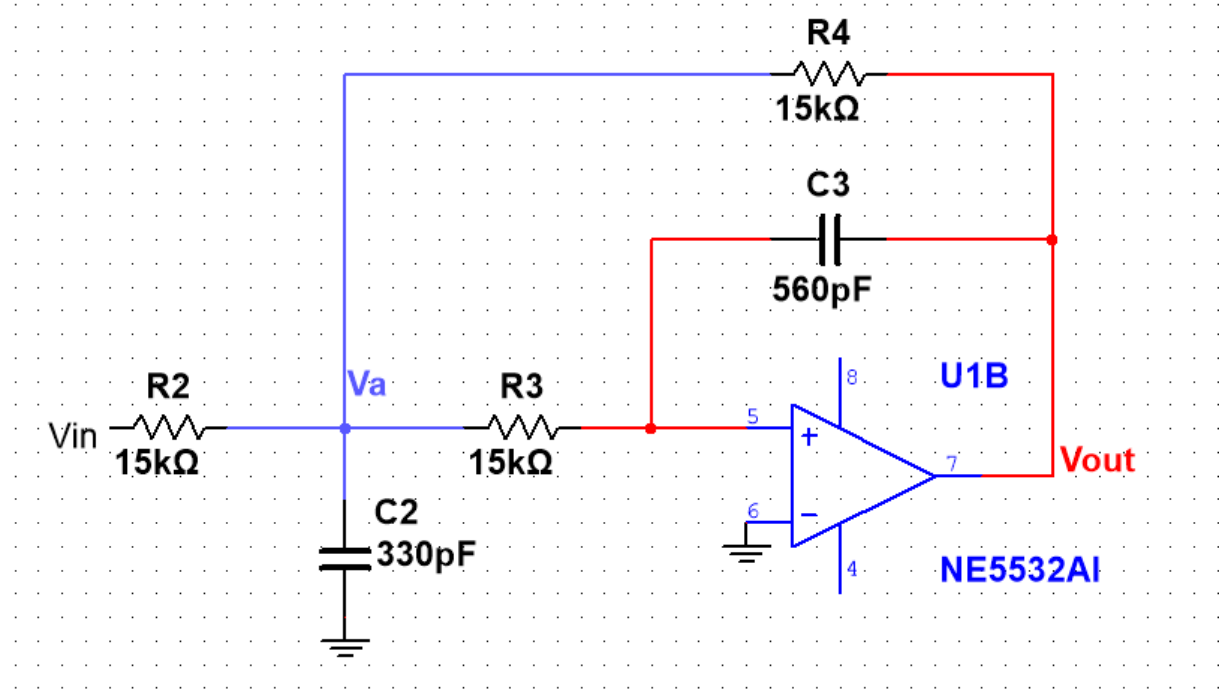


Figure 17: Multiple-feedback low-pass filter topology.

The inverting op-amp configuration gives the following relation between V_a and V_{out} :

$$v_{out} = \frac{-Z_2 V_a}{Z_1} = \frac{-\frac{1}{sC_3} V_a}{R_3},$$

$$v_a = -v_{out} R_3 C_3 s.$$

Kirchhoff's current law can then be applied at the node with voltage V_a :

$$I_{R2} + I_{R3} + I_{C2} + I_{C4} = \frac{V_{in} - V_a}{R_2} + \frac{0 - V_a}{\frac{1}{C_2 s}} + \frac{0 - V_a}{R_3} + \frac{V_{out} - V_a}{R_4} = 0.$$

Substituting the expression for V_a and rearranging yields

$$T(s) = \frac{-\frac{1}{R_2 R_3 C_2 C_3}}{s^2 + s\left(\frac{1}{R_2 C_2} + \frac{1}{R_3 C_2} + \frac{1}{R_4 C_2}\right) + \frac{1}{R_3 R_4 C_2 C_3}}.$$

Comparison with the standard second-order low-pass form gives

$$\omega_0 = \sqrt{\frac{1}{R_3 R_4 C_2 C_3}}, \quad Q = \frac{\sqrt{R_3 R_4 C_2 C_3}}{\frac{1}{R_2 C_2} + \frac{1}{R_3 C_2} + \frac{1}{R_4 C_2}}, \quad K = -\frac{R_4}{R_2}.$$

8.2 PT2399 configuration

The PT2399 configuration was based mainly on its datasheet, independent manufacturers' designs, and forum discussions about the integrated circuit. The most relevant sources were:

- PT2399 Datasheet – Electric Druid
- PT2399 Analysis – Electrosplash
- Boy in Well – DIY Guitar Pedals